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**Table 4.1 Properties of Hydrogen, Deuterium and Tritium molecules.**

| Property | Protium | Deuterium | Tritium |
|---|---|---|---|
| Atomic Hydrogen | | | |
| | H | D | T |
| Atomic mass (*amu*) | 1.008 | 2.014 | 3.016 |
| No of electron / protons / neutrons | 1 / 1 / 0 | 1 / 1 / 1 | 1 / 1 / 2 |
| Nuclear stability | Stable | Stable | Radioactive ($t_{1/2}$=12.3 yrs) |
| Molecular Hydrogen | $H_2$ | $D_2$ | $T_2$ |
| Abundance (%) | 99.985 | 0.015 | $\sim 10^{-16}$ |
| Molecular mass (*amu*) | 2.016 | 4.028 | 6.032 |
| Melting point (*K*) | 13.96 | 18.73 | 20.62 |
| Boiling point (*K*) | 20.30 | 23.67 | 25.04 |
| Internuclear distance (*pm*) | 74.14 | 74.14 | 74.14 |
| Critical Temperature (*K*) | 33.19 | 38.35 | 40.60 |
| Enthalpy of dissociation (k*J/mol*)† | 435.9 | 443.4 | 446.9 |

† Bond dissociation energy

### 4.1.3 Ortho– and Para–Hydrogen:

In the hydrogen atom, the nucleus has a spin. When molecular hydrogen is formed, the spins of two hydrogen nuclei can be in the same direction or in the opposite direction as shown in the figure. These two forms of hydrogen molecules are called *ortho* and *para* hydrogens respectively.



Ortho-hydrogen Para-hydrogen

At room temperature, normal hydrogen consists of about 75% ortho-form and 25% para-form. As the ortho-form is more stable than para-form, the conversion of one isomer into the other is a slow process. However, the equilibrium shifts in favour of para hydrogen when the temperature is lowered. The para-form can be catalytically transformed into ortho-form using platinum or iron. Alternatively, it can also be converted by passing an electric discharge, heating above 800°C and mixing with paramagnetic molecules such as $O_2$, NO, $NO_2$ or with nascent/atomic hydrogen.